Michigan Tech

The Matrix Exponential

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$\vec{y}' = A\vec{y}$: The Matrix Exponential e^A

For a square matrix A, the matrix exponential is defined by

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

If $A = V\Lambda V^{-1}$ then

$$e^A = Ve^{\Lambda}V^{-1} = V \begin{pmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{pmatrix} V^{-1}, \qquad \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

Review: Three power series

Read down each column to compare terms of the same degree.

$$\begin{array}{rccccccc} & x^0 & x^1 & x^2 & x^3 & x^4 & x^5 & \\ \hline e^x = & 1 & + x + & \frac{x^2}{2!} & + \frac{x^3}{3!} & + \frac{x^4}{4!} & + \frac{x^5}{5!} & + \cdots \\ \cos x = & 1 & + 0 - & \frac{x^2}{2!} & + 0 & + \frac{x^4}{4!} & + 0 & + \cdots \\ \sin x = & 0 & + x + 0 - & \frac{x^3}{3!} & + 0 & + \frac{x^5}{5!} & & + \cdots \end{array}$$

Cosine has only even powers. Sine has only odd powers.

Warm-up: Euler's formula

For real y, substitute $x = iy$ into the exponential series.

$$\begin{aligned} e^{iy} &= 1 + iy + \frac{(iy)^2}{2!} + \frac{(iy)^3}{3!} + \frac{(iy)^4}{4!} + \frac{(iy)^5}{5!} + \cdots \\ &= 1 + iy - \frac{y^2}{2!} - i\frac{y^3}{3!} + \frac{y^4}{4!} + i\frac{y^5}{5!} + \cdots \\ &= \left(1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \cdots\right) + i\left(y - \frac{y^3}{3!} + \frac{y^5}{5!} - \cdots\right) \\ &= \cos y + i \sin y. \end{aligned}$$

The even powers are real and the odd powers carry a factor of i.

Why the matrix exponential is a fundamental matrix

A *fundamental matrix* for $\mathbf{y}' = A\mathbf{y}$ is an invertible matrix $\Phi(t)$ whose columns solve the system. Equivalently, $\Phi' = A\Phi$ and its columns are linearly independent.
For $\Phi(t) = e^{tA}$, differentiate the power series term by term (it converges uniformly on bounded intervals, as does its derivative):

$$\Phi'(t) = \sum_{n=1}^{\infty} \frac{t^{n-1}A^n}{(n-1)!} = A\sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = A\Phi(t).$$

Also, $\Phi(0) = I$ and $e^{tA}e^{-tA} = I$ by multiplying the series. Thus e^{tA} is the fundamental matrix normalized at $t = 0$.

Example: Split the real pseudodiagonal form into blocks

Suppose $A = VBV^{-1}$, with V real and invertible, and write

$$B = \begin{pmatrix} a & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & -\beta & \alpha \end{pmatrix} = \operatorname{diag}(a, C), \qquad C = \alpha I + \beta J, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where I is the 2×2 identity and $J^2 = -I$. Since $IJ = JI = J$, the binomial theorem applies. Using $J^2 = -I$ and $J^3 = -J$,

$$C^2 = (\alpha I + \beta J)^2 = \alpha^2 I + 2\alpha\beta J + \beta^2 J^2 = (\alpha^2 - \beta^2) I + 2\alpha\beta J,$$
$$C^3 = (\alpha I + \beta J)^3 = \alpha^3 I + 3\alpha^2\beta J + 3\alpha\beta^2 J^2 + \beta^3 J^3 = (\alpha^3 - 3\alpha\beta^2) I + (3\alpha^2\beta - \beta^3) J.$$

$$\vdots$$

Example: Compute the eigenvalues of B

Expanding the determinant along the first row gives

$$p_B(\lambda) = \det(\lambda I - B) = \det \begin{pmatrix} \lambda - a & 0 & 0 \\ 0 & \lambda - \alpha & -\beta \\ 0 & \beta & \lambda - \alpha \end{pmatrix}$$

$$= (\lambda - a) \det \begin{pmatrix} \lambda - \alpha & -\beta \\ \beta & \lambda - \alpha \end{pmatrix}$$

$$= (\lambda - a)\big((\lambda - \alpha)^2 + \beta^2\big).$$

The first factor gives $\lambda = a$. The second gives $(\lambda - \alpha)^2 = -\beta^2$, so $\lambda = \alpha \pm i\beta$.

Example: Factor the power series

In the nth power, pick m factors of $\beta t J$ and $n-m$ factors of $\alpha t I$. Since $\binom{n}{m}/n! = 1/((n-m)!\,m!)$,

$$e^{tC} = \sum_{n=0}^{\infty} \sum_{m=0}^{n} \frac{(\alpha t)^{n-m}}{(n-m)!} \frac{(\beta t)^m J^m}{m!}$$
$$= \left(\sum_{k=0}^{\infty} \frac{(\alpha t)^k}{k!} I \right) \left(\sum_{m=0}^{\infty} \frac{(\beta t)^m}{m!} J^m \right).$$

Reindex with $k = n-m$ and regroup the absolutely convergent series. The first factor is $(1 + \alpha t + (\alpha t)^2/2! + \cdots)I = e^{\alpha t} I$.

Example: The even and odd powers give cosine and sine

The powers cycle as $I, J, -I, -J, I, \ldots$ since $J^2 = -I$. Thus $J^{2k} = (-1)^k I$ and $J^{2k+1} = (-1)^k J$. Splitting the second series into even and odd powers gives

$$\begin{aligned} e^{\beta t J} &= \left(1 - \frac{(\beta t)^2}{2!} + \frac{(\beta t)^4}{4!} - \cdots\right) I \\ &\quad + \left(\beta t - \frac{(\beta t)^3}{3!} + \frac{(\beta t)^5}{5!} - \cdots\right) J \\ &= \cos(\beta t) I + \sin(\beta t) J. \end{aligned}$$

Multiplying by $e^{\alpha t} I$ gives

$$e^{tC} = e^{\alpha t} \begin{pmatrix} \cos(\beta t) & \sin(\beta t) \\ -\sin(\beta t) & \cos(\beta t) \end{pmatrix}.$$

Example: Assemble the blocks and change basis

Powers of B inherit the block structure.

- $B^2 = \begin{pmatrix} a^2 & 0 \\ 0 & C^2 \end{pmatrix}, \qquad B^3 = \begin{pmatrix} a^3 & 0 \\ 0 & C^3 \end{pmatrix}, \qquad \dots$
- The scalar block is $\sum_{n=0}^{\infty} (at)^n/n! = e^{at}$.
- The lower block is e^{tC}.

Since $A^n = VB^nV^{-1}$, the power series is

$$e^{tA} = Ve^{tB}V^{-1} = V \begin{pmatrix} e^{at} & 0 & 0 \\ 0 & e^{\alpha t}\cos(\beta t) & e^{\alpha t}\sin(\beta t) \\ 0 & -e^{\alpha t}\sin(\beta t) & e^{\alpha t}\cos(\beta t) \end{pmatrix} V^{-1}.$$